

Úloha 2

a) Provedeme Gram-Schmidtovu ortogonalizaci pro posloupnost (v_1, v_2, v_3) , kde $v_1 = 1, v_2 = x, v_3 = x^2$

$$\vec{u}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|} = \frac{1}{\sqrt{\int_1^5 1 \cdot 1 dx}} = \frac{1}{\sqrt{[x]_1^5}} = \frac{1}{\sqrt{5-1}} = \frac{1}{2}$$

$$\vec{w}_2 = \int_1^5 \vec{u}_1 \cdot \vec{v}_2 dx \cdot \vec{u}_1 = \int_1^5 \frac{1}{2} \cdot x dx \cdot \frac{1}{2} = \frac{1}{2} \left[\frac{x^2}{2} \right]_1^5 \cdot \frac{1}{2} = \frac{1}{4} \left(\frac{25}{2} - \frac{1}{2} \right) = \frac{1}{4} \cdot 12 = 3$$

$$\vec{u}_2 = \frac{\vec{v}_2 - \vec{w}_2}{\sqrt{\int_1^5 (\vec{v}_2 - \vec{w}_2)^2 dx}} = \frac{x-3}{\sqrt{\int_1^5 (x-3)^2 dx}} = \frac{x-3}{\sqrt{\frac{16}{3}}} = \frac{x-3}{\frac{4}{\sqrt{3}}} = \frac{\sqrt{3}}{4} \cdot (x-3)$$

$$\vec{w}_3 = \int_1^5 \vec{u}_1 \vec{v}_3 dx \cdot \vec{u}_1 + \int_1^5 \vec{u}_2 \vec{v}_3 dx \cdot \vec{u}_2 = \int_1^5 \frac{1}{2} \cdot x^2 dx \cdot \frac{1}{2} + \int_1^5 \frac{\sqrt{3}}{4} (x-3) \cdot x^2 dx \cdot \frac{\sqrt{3}}{4} (x-3) = \frac{62}{3} \cdot \frac{1}{2} + 8\sqrt{3} \cdot \frac{5}{4} (x-3) = \frac{31}{3} + 6(x-3) =$$

$$= 6x - \frac{23}{3}$$

$$\vec{u}_3 = \frac{\vec{v}_3 - \vec{w}_3}{\sqrt{\int_1^5 (\vec{v}_3 - \vec{w}_3)^2 dx}} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx}} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\frac{256}{45}}} = \frac{x^2 - 6x + \frac{23}{3}}{\frac{16}{\sqrt{45}}} = \frac{3\sqrt{5}}{16} \cdot (x^2 - 6x + \frac{23}{3})$$

Ortonormální báze podprostoru generovaného $\text{LO}[1, x, x^2]$ je tedy $B = \left(\left(\frac{1}{2} \right), \left(\frac{\sqrt{3}}{4} (x-3) \right), \left(\frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}) \right) \right)$

b) hledáme ortogonální projekci fce $\sin(x)$ na tento podprostor vyjádřenou jako $ax^2 + bx + c \rightarrow$ vlastně hledáme nejblížeší aproximaci fce $\sin(x)$ vzhledem k vypočítané ortonormální bázi

$$(\sin(x))_{[B]} = \begin{pmatrix} \langle \vec{u}_1, \sin x \rangle \\ \langle \vec{u}_2, \sin x \rangle \\ \langle \vec{u}_3, \sin x \rangle \end{pmatrix} \Rightarrow ax^2 + bx + c = \langle \vec{u}_1, \sin x \rangle \vec{u}_1 + \langle \vec{u}_2, \sin x \rangle \vec{u}_2 + \langle \vec{u}_3, \sin x \rangle \vec{u}_3$$

$$ax^2 + bx + c = \int_1^5 \frac{1}{2} \sin x dx \cdot \frac{1}{2} + \int_1^5 \frac{\sqrt{3}}{4} (x-3) \sin x dx \cdot \frac{\sqrt{3}}{4} (x-3) + \int_1^5 \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}) \sin x dx \cdot \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}) =$$

$$= 0,12832 \cdot \frac{1}{2} - 1,4932 \cdot \frac{\sqrt{3}}{4} (x-3) - 0,12524 \cdot \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}) = -0,0525x^2 - 0,3315x + 1,6039$$

Koeficienty: $a = -0,0525$ $b = -0,3315$, $c = 1,6039$

$$c) \|\sin x - (ax^2 + bx + c)\| = \sqrt{\int_1^5 (\sin x - ax^2 - bx - c)^2 dx} = \sqrt{\int_1^5 (\sin x + 0,0525x^2 + 0,3315x - 1,6039)^2 dx} = 0,318837$$